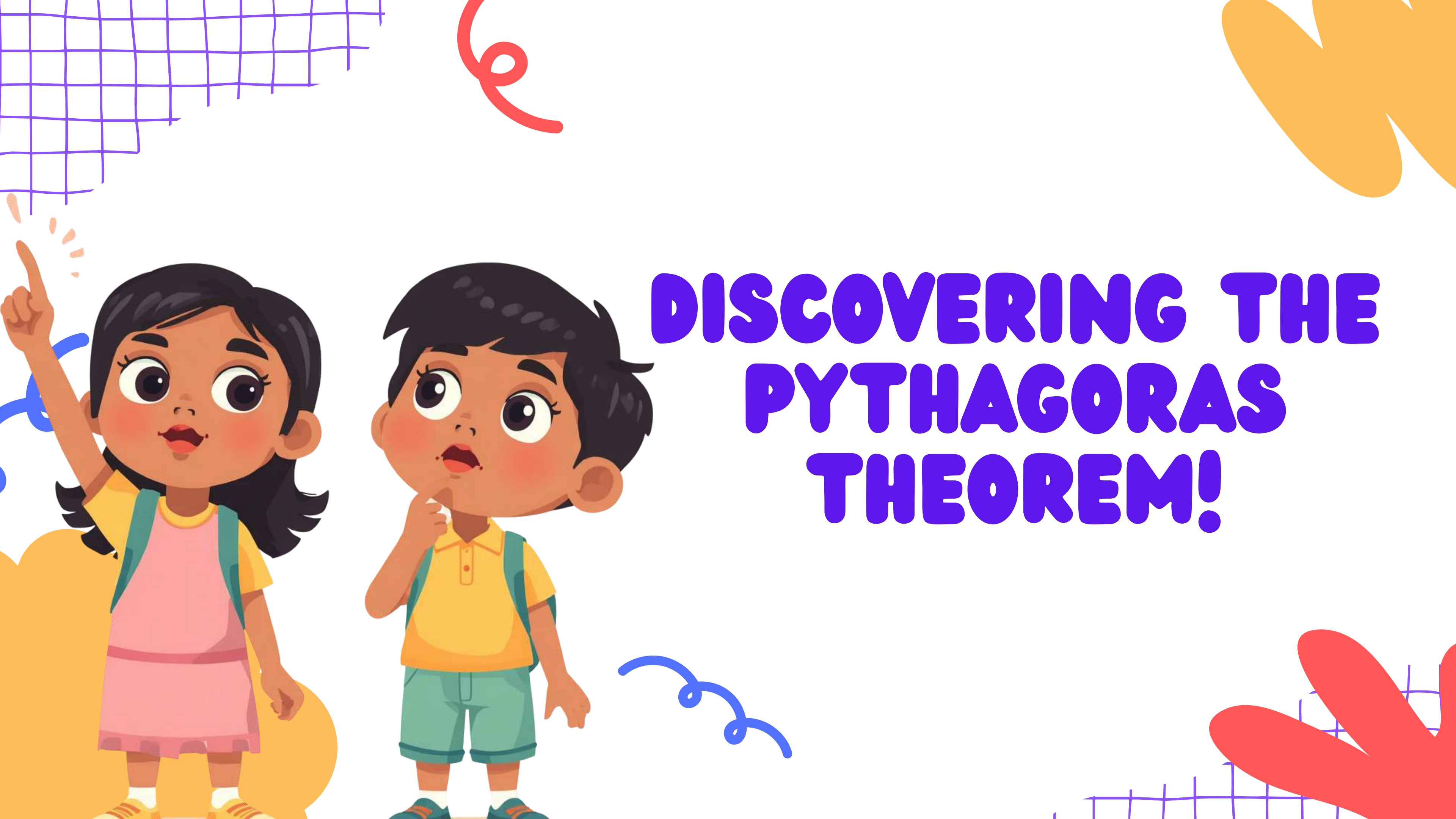




# DISCOVERING THE PYTHAGORAS THEOREM!

# MEET OUR SHAPES FRIENDS!

Let's explore some shapes and their special secrets! Triangles, squares, and rectangles are all around us – and they hold amazing maths magic!

Spot the triangle, square, and rectangle around you!

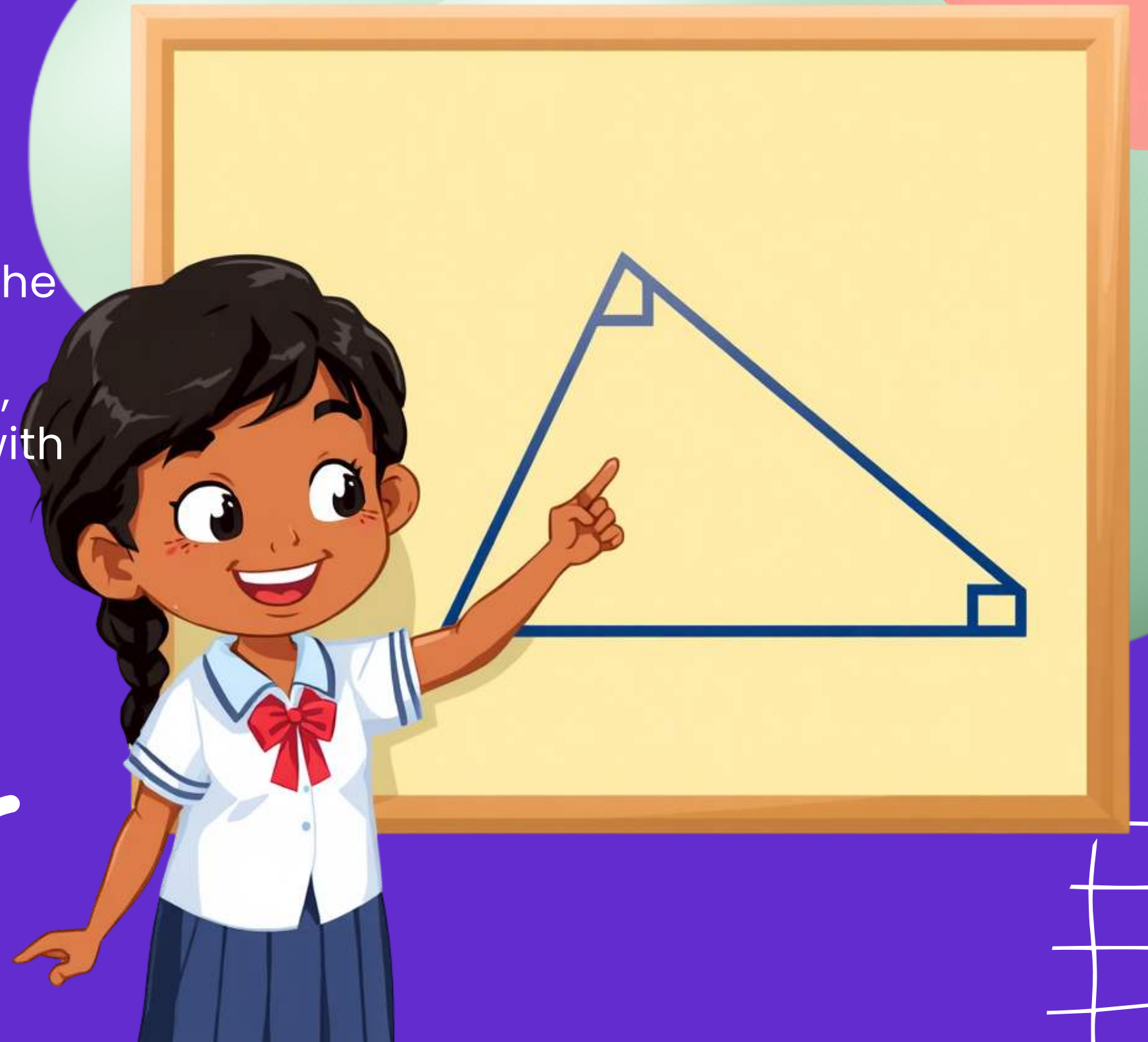
Every shape has sides, corners, and a secret to share!



# WHAT IS A RIGHT-ANGLED TRIANGLE?

A right-angled triangle has one special corner — just like the corner of a book or the edge of your notebook! That special corner is called a right angle, and it measures exactly  $90^\circ$ . We mark it with a tiny square symbol in the corner.

**Look for the little square — that's your right angle!**





# LET'S LOOK AT THE SIDES!

Every right-angled triangle has three special sides. Each side has its own name and role. Let's learn what makes each side unique!

Hypotenuse: The longest side, always opposite the right angle.

Side A & Side B: The two shorter sides that form the right angle.

Tip: The hypotenuse is always the biggest side in a right triangle!



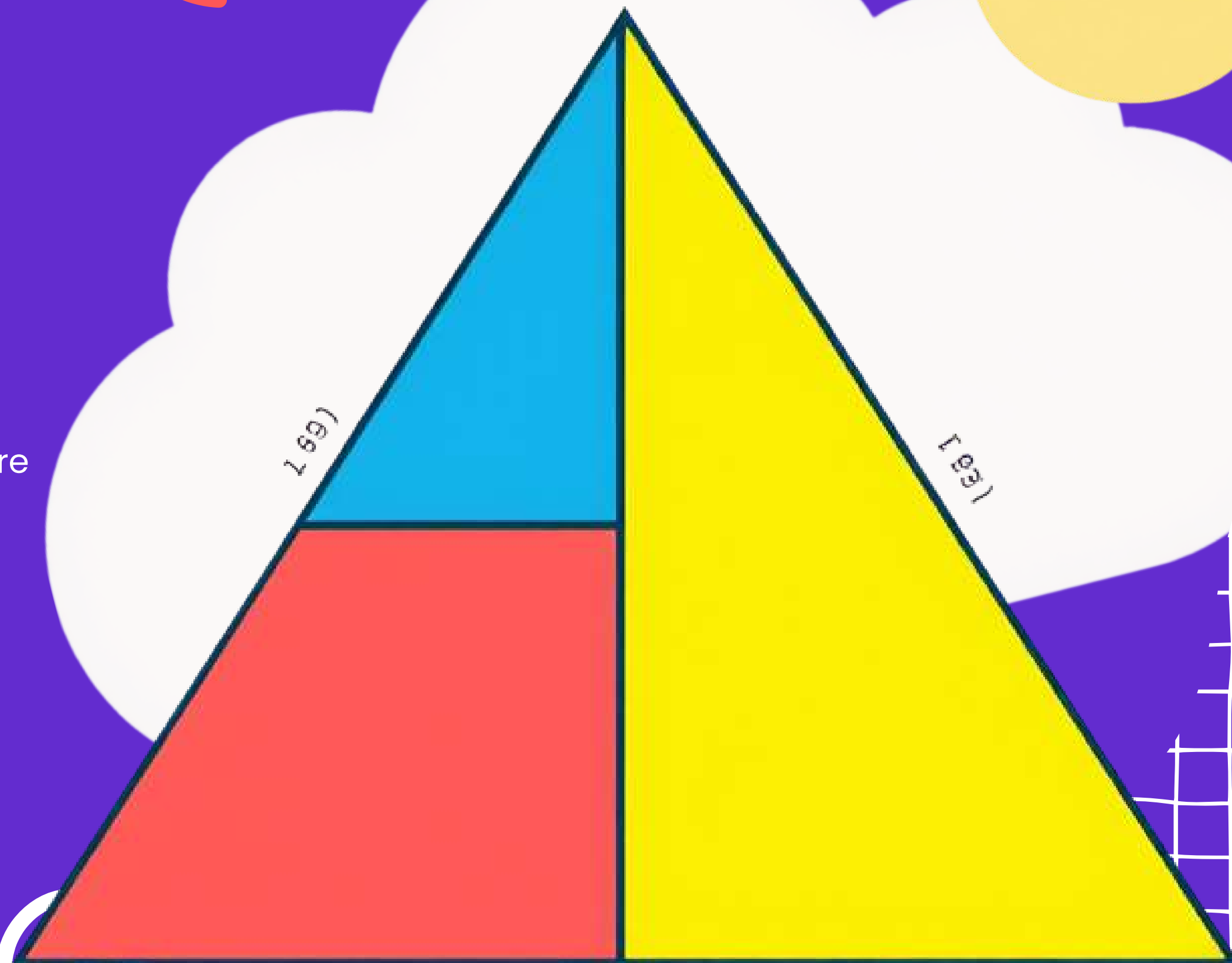
# SQUARES ON EACH SIDE!

## Building Squares

Look! We made squares on each side of the triangle. Each side has its own colourful square – can you see how they fit together?



**The squares on Side A and Side B together equal the square on the Hypotenuse!**

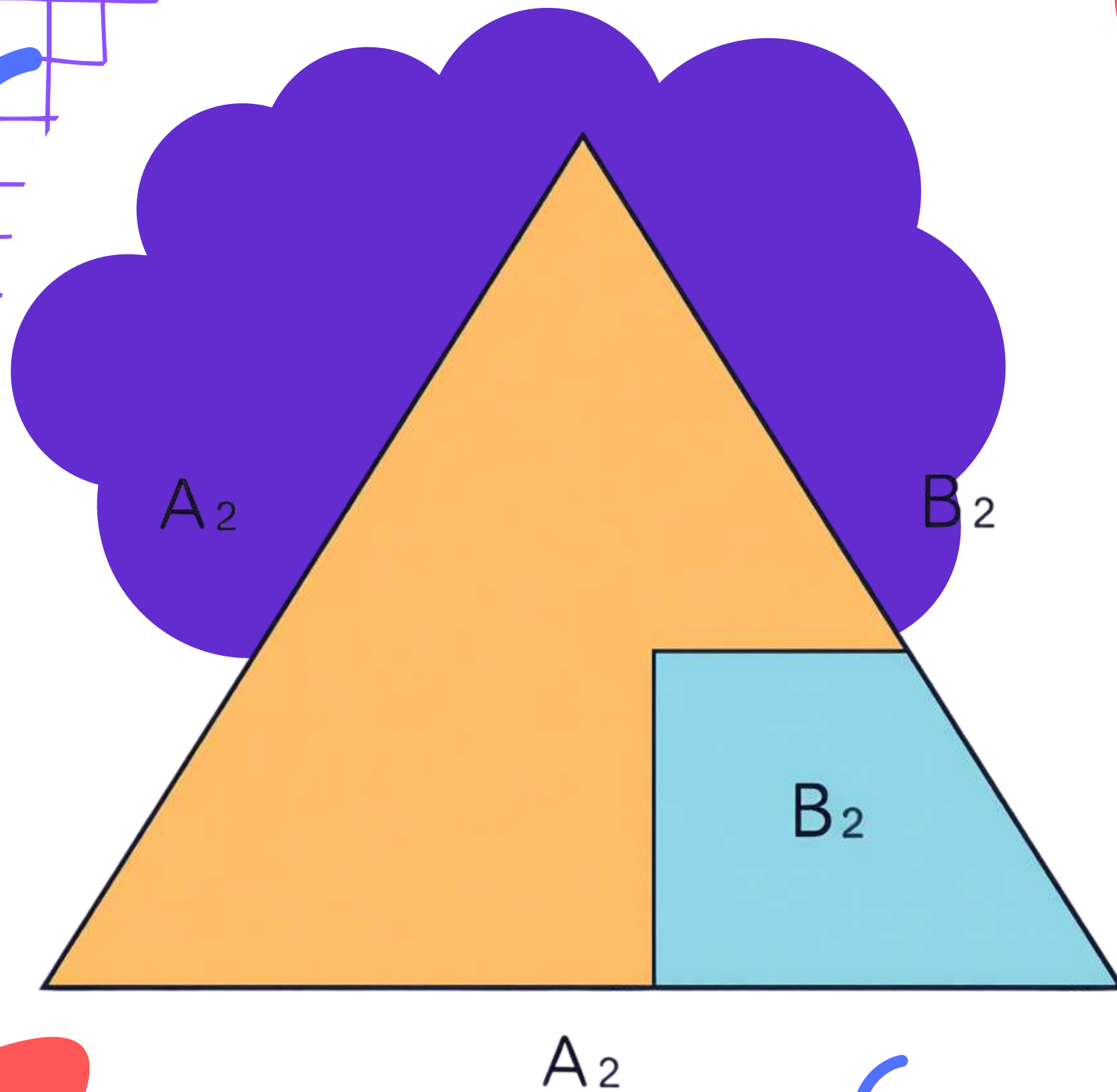


# WHAT DOES THE THEOREM SAY?

The area of the big square equals the  
sum of the areas of the two smaller  
squares. In short:  
 $a^2 + b^2 = c^2$

$c^2$  is the biggest square — on the  
longest side (hypotenuse)!

$a^2 + b^2 = c^2 \rightarrow$  Add the two small  
squares to get the big one!



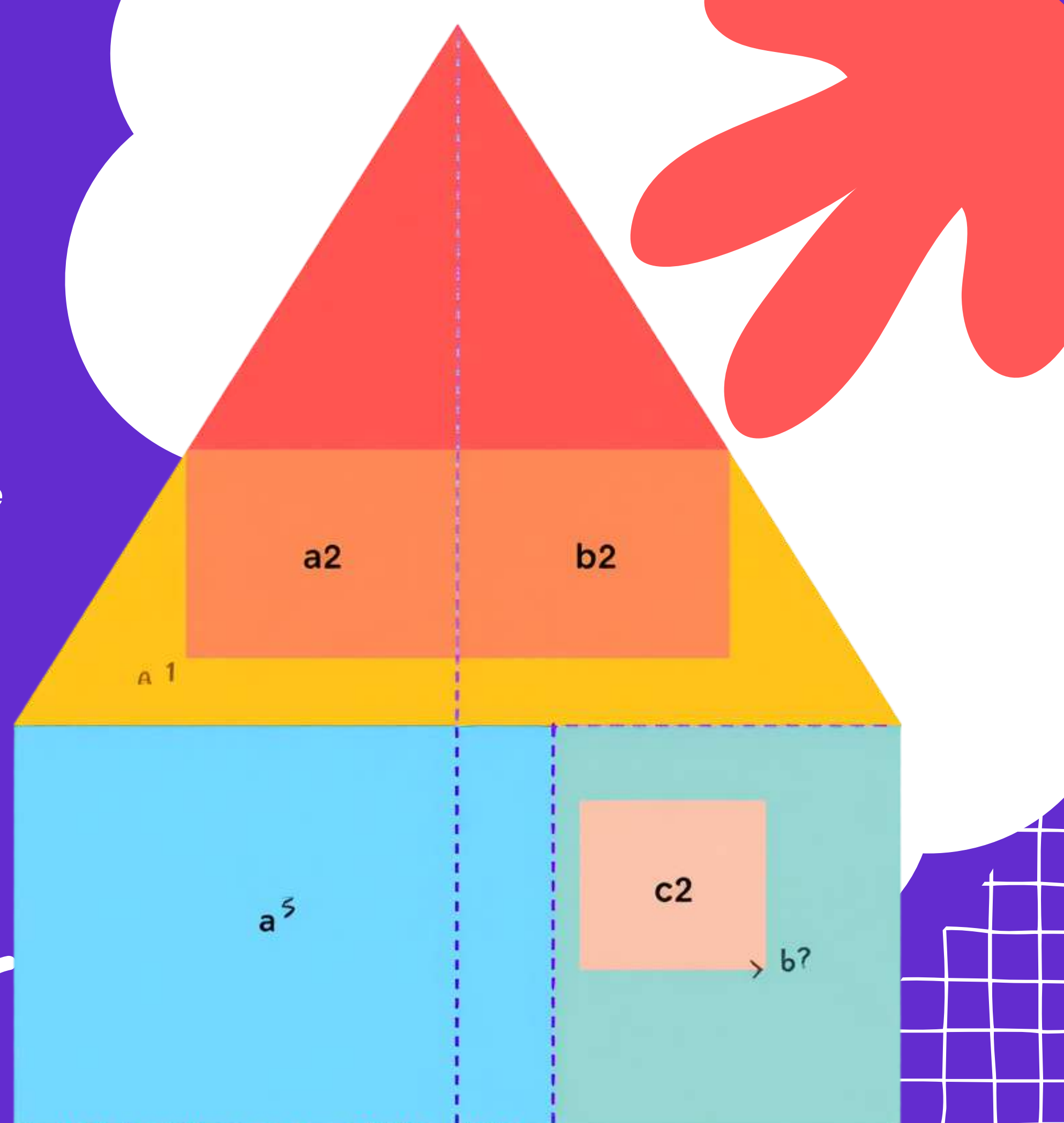
# VISUAL PROOF — LET'S SEE!

See how the two small squares fit into the big one?

The square on the longest side ( $c^2$ ) has exactly the same area as the two smaller squares ( $a^2$  and  $b^2$ ) added together!

$$a^2 + b^2 = c^2$$

**The big square = the two small squares combined.  
Every time!**





# REAL-LIFE EXAMPLE: THE LADDER PROBLEM!

A ladder leans against a wall and forms a right-angled triangle! How far up the wall can the ladder reach? Let's use the Pythagoras theorem to find out.

The ladder is the hypotenuse — the longest side of the triangle!

Use  $a^2 + b^2 = c^2$  to find the missing side length.





# CLASSROOM EXAMPLE

## Measuring on the Floor!

Riya and Arjun mark a right triangle on the classroom floor. One side is 3 metres long and the other is 4 metres. Can you help them find the length of the longest side?

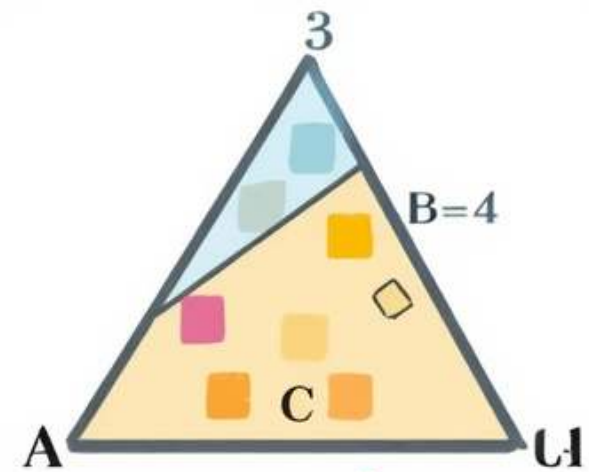


**What do we need to find?**  
- The hypotenuse — the longest side of the triangle!



# LET'S CALCULATE TOGETHER!

Let's solve a triangle step by step! If  $a = 3$  units and  $b = 4$  units, we can find the longest side  $c$  using  $a^2 + b^2 = c^2$



Step 1:

$$a^2 = 3^2 = 9$$
$$b^2 = 4^2 = 16$$

Step 2:

$$a^2 + b^2 = c^2$$
$$9 + 16 = 25$$

Step 3:

$$c^2 = 25$$
$$\text{So } c = \sqrt{25}$$
$$c = 5 \text{ units! } \checkmark$$

Answer: The hypotenuse is 5 units long!

# PRACTICE PROBLEMS!

Try these on your own — use  $a^2 + b^2 = c^2$  to find the answers.  
You've got this!

Find the hypotenuse when  $a = 6$  and  $b = 8$ .

Find the missing side when  $c = 10$  and  $a = 6$ .

True or False: A triangle with sides 5, 12, 13 is right-angled?





# FUN QUIZ TIME!

Which side is the hypotenuse? Look at the triangles below and think carefully before you answer!

Triangle A: Sides 3, 4, 5 —  
Which is the longest side?

Triangle B: Sides 6, 8, 10 — Can you find it?

Triangle C: Sides 5, 12, 13 — Point to the hypotenuse!

Hint: The hypotenuse is always the side opposite the right angle!







# THANK YOU FOR LEARNING WITH US!

**You Did Amazing Today!**

You explored right-angled triangles, met the hypotenuse, and discovered the magic of  $a^2 + b^2 = c^2$ ! Keep practising and you will be a Pythagoras pro!

**Keep curious, keep exploring — maths  
is full of wonderful surprises!**